Object Oriented Programming

Java

Introduction To Java

- The most widely used computer programming language.
- Specifically designed to have as few implementation dependencies as possible.
- Java was originally developed by *James Gosling* at Sun Microsystems and now it has been acquired by Oracle Corporation.
- *James Gosling* designed Java with a C/C++ style syntax that system and application programmers would find familiar.
- The language was initially called Oak after an oak tree that stood outside Gosling's office.
- Later the project went by the name Green and was finally renamed Java, from Java coffee.

Introduction To Java

- Three editions:
  1. Standard Edition (SE)
  2. Enterprise Edition (EE)
     - Large-scale, distributed networking and web-based applications.
  3. Micro Edition (ME)
     - Developing applications for small, memory-constrained devices, such as BlackBerry smartphones.

Java Development Environment

- A key goal of Java is to be able to write programs that will run on a great variety of computer systems and computer-control devices. This is sometimes called "write once, run anywhere."
- ***Java Class Libraries:*** Java programmers take advantage of the rich collections of existing classes and methods in the **Java class libraries**, which are also known as the **Java APIs (Application Programming Interfaces)**.
- Java programs normally go through five phases—edit, compile, load, verify and execute.
- Java SE Development Kit (JDK)

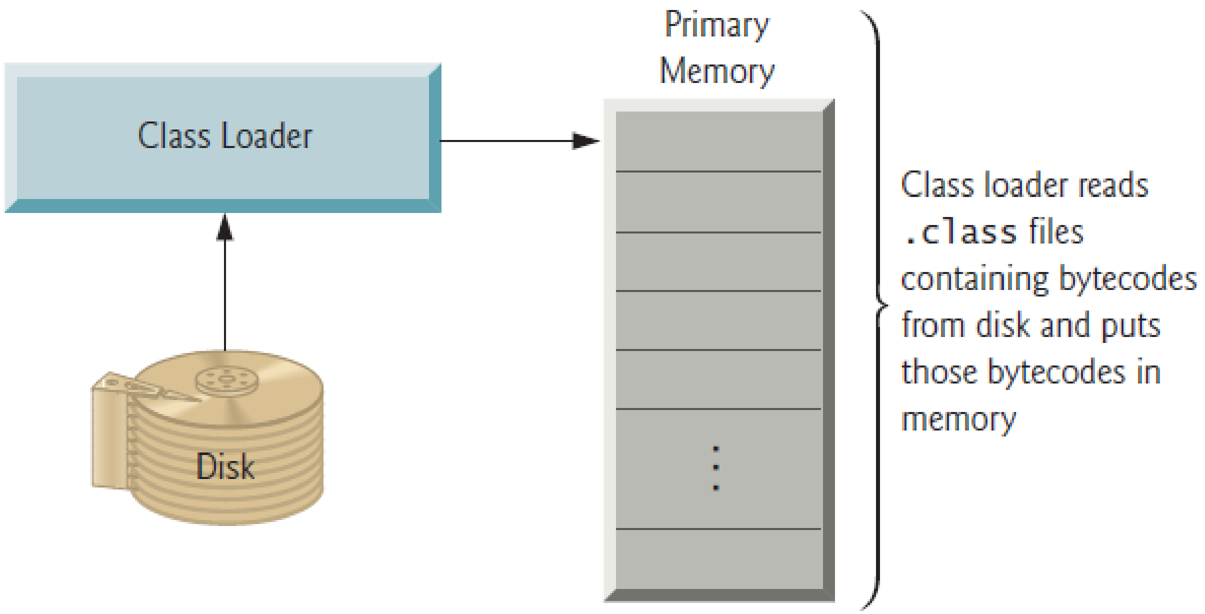
Java Development Environment

- ***Phase 1: Creating a Program***
  - editing a file with an *editor program*
  - A file name ending with the ***.java* extension** indicates that the file contains Java source code.
- ***Phase 2: Compiling a Java Program into Bytecodes***
  - using the command **javac** (the **Java compiler**) to **compile** a program
  - If the program compiles, the compiler produces a **.class** file called *Welcome.class* that contains the compiled version of the program.
  - The Java compiler translates Java source code into **bytecodes**

N. B.: Bytecodes are platform independent—they do not depend on a particular hardware platform. So, Java's bytecodes are **portable.**

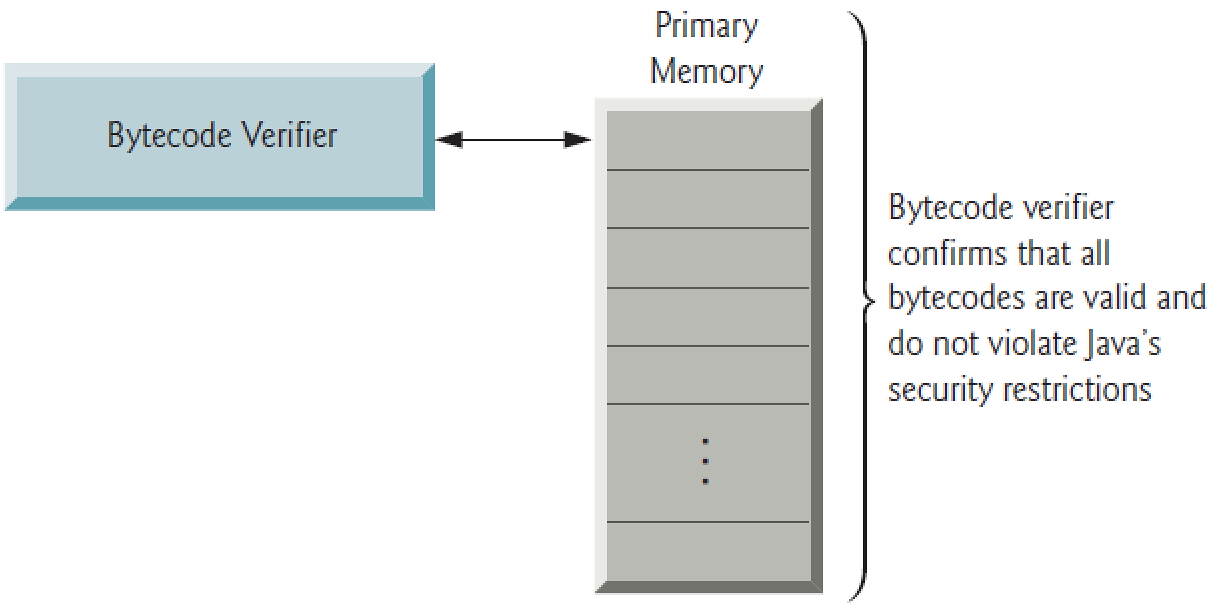
Java Development Environment

- ***Phase 3: Loading a Program into Memory***
  - Java Virtual Machine (JVM) places the program in memory to execute it—this is known as **loading**
  - The JVM's **class loader** takes the *.class* files containing the program's bytecodes and transfers them to primary memory.



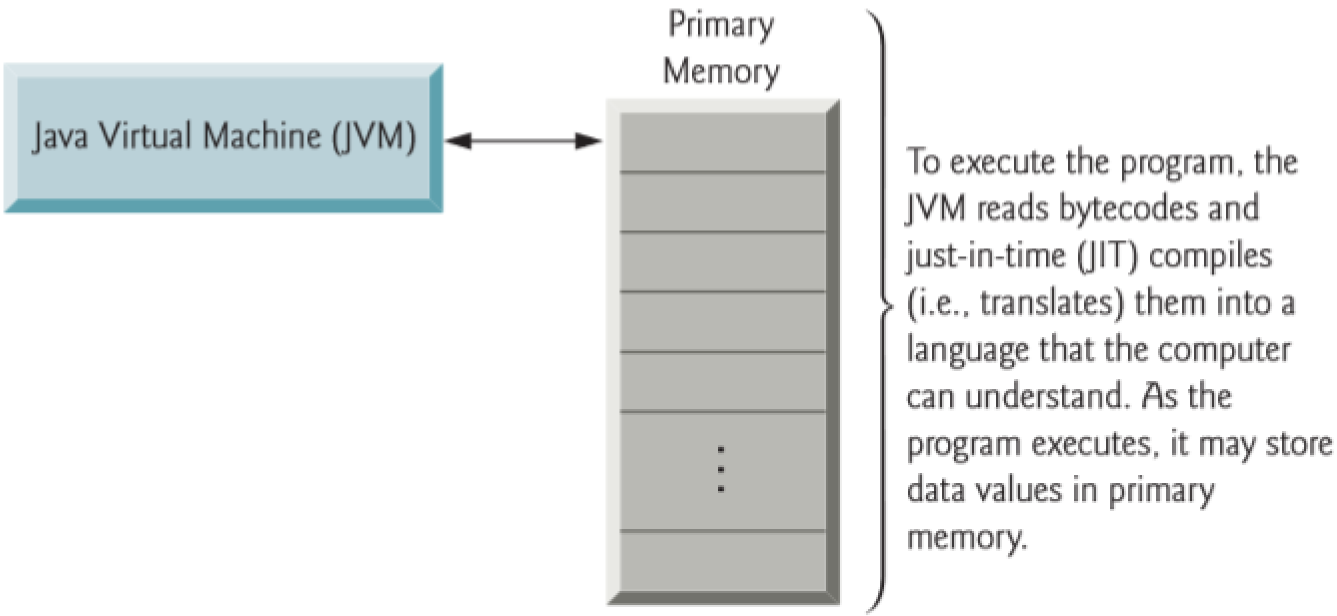
Java Development Environment

- ***Phase 4: Bytecode Verification***
  - As the classes are loaded, the **bytecode verifier** examines their bytecodes to ensure that they're valid and do not violate Java's security restrictions.
  - Java enforces strong security to make sure that Java programs arriving over the network do not damage your files or your system (as computer viruses and worms might).



Java Development Environment

- ***Phase 5: Execution***
  - JVM executes the program’s bytecodes, thus performing the actions specified by the program.



Java Development Environment

- **The Java Development Kit (JDK)** is a software development environment used for developing Java applications and applets.
  - It includes the Java Runtime Environment (JRE), an interpreter/loader (java), a compiler (javac), an archiver (jar), a documentation generator (javadoc) and other tools needed in Java development.